

WREN

Wear with intention.

Feature Specification Document

All Core Features · v2.0 · MVP + V1 + V2 Scope

Document Type: Feature Specification Document

App Name: WREN — AI-Powered Personal Wardrobe App

Version: 2.0

Status: MVP Development Phase

Scope: 14 Core Features (FS-01 to FS-14)

Platform: iOS & Android (React Native + Expo)

Audience: Engineering, Design, QA, Academic Evaluation

How to read this document: Each feature spec (FS-01 to FS-14) is self-contained and covers: overview, user stories, functional requirements, acceptance criteria, edge cases, UI specification, API contract, data model, testing requirements, and success metrics. Features are ordered by MoSCoW priority then dependency. Read sequentially for full context or jump to any FS-XX for individual feature detail.

Feature Index

FS ID	Feature Name	Priority	Personas	Release
FS-01	Wardrobe Digitization & AI Scan	M — Must	All	MVP
FS-02	AI Outfit Generation Engine	M — Must	Kofi, Amara	MVP
FS-03	Style Profile & Onboarding Quiz	M — Must	All	MVP
FS-04	Home Screen — Morning Brief	M — Must	Amara, Kofi	MVP
FS-05	Wear Logging & Post-Wear Rating	M — Must	All	MVP
FS-06	Bottom Tab Navigator	M — Must	All	MVP
FS-07	Style Feed — Masonry Grid	S — Should	Kofi, Esi	V1
FS-08	Wardrobe Gallery — The Studio	S — Should	All	V1
FS-09	Outfit Recommendations Screen	S — Should	Amara, Kofi	V1
FS-10	Virtual Try-On — Flat-Lay Studio	S — Should	Kofi, Esi	V1
FS-11	Weekly Outfit Planner	S — Should	Amara	V1
FS-12	Outfit Collections	S — Should	Kofi	V1
FS-13	Monthly Wrapped	C — Could	All	V1
FS-14	On-Body Virtual Try-On	C — Could	Kofi, Esi	V2

1.1 Feature Overview

Field	Detail
Problem Statement	A digital wardrobe with no intelligence is just a photo gallery. But a wardrobe that cannot be built quickly will never be built at all. The scan experience is WREN's first impression and the gateway to every other feature.
Solution	A guided camera flow with AI auto-tagging that takes a clothing item from photo to fully catalogued wardrobe entry in under 90 seconds. Google Cloud Vision API provides category, colour, pattern and material detection. All tags are editable chips — user corrections are stored and used for model retraining.
Key Differentiator	The scan is not a form. It is a reveal. Tags animate onto the screen with spring physics after scan, making AI inference feel like the app is thinking, not filling in a spreadsheet. Confidence gating means the AI only asserts what it knows; it prompts the user for everything else.
Design Reference	Airbnb photo capture guidance (lighting cues, frame overlay) × Calm onboarding (one action per screen, no cognitive overload)

1.2 User Stories

Story 1 — Single Item Scan

As Amara (busy professional), I want to scan a clothing item and have it automatically tagged with its category, colour, and key attributes, so I can build my wardrobe library in minutes rather than hours.

Story 2 — Tag Correction

As Kofi (fashion enthusiast), I want to correct any inaccurate AI tags immediately after scanning, so my wardrobe data stays precise and my outfit suggestions stay relevant.

Story 3 — Batch Scan

As Esi (minimalist student), I want to scan multiple items in one session without being sent back to the home screen between each item, so I can digitize my entire wardrobe in one sitting.

1.3 Functional Requirements

Camera & Image Capture

- **FR-01-01 Launch camera with full-screen viewfinder and white corner-bracket overlay indicating garment framing zone**
- FR-01-02 Real-time lighting quality indicator: green dot (good), amber (borderline), red (too dark) — updates at 500ms intervals
- FR-01-03 Detect blur in real time via Laplacian variance; prevent capture if score below threshold
- FR-01-04 Support photo capture (primary) and photo library upload (secondary — "Choose from library" CTA)
- FR-01-05 Capture at minimum 1080px on the shorter dimension; resize to 800px for API call, store original at 1080px

AI Processing Pipeline

- **FR-01-06 On capture: run background removal using remove.bg API or equivalent; deliver isolated garment PNG**
- FR-01-07 Submit to Google Cloud Vision API; receive and parse clothing labels, dominant colours (hex), pattern, logo detection
- FR-01-08 Confidence gating: category confidence <60% → disambiguation prompt; brand confidence <50% → set brand = "Unknown"
- FR-01-09 Map Vision API labels to WREN category taxonomy: tops, bottoms, dresses, outerwear, footwear, accessories
- FR-01-10 Assign formality score 1–4 from category + label combination (e.g. "Oxford shirt" → 3, "T-shirt" → 1)
- FR-01-11 Generate thumbnail at 400x400px for gallery grid display; store separately from full image

Tag Review Screen

- **FR-01-12 Display all AI-generated tags as editable chips on review screen; user can tap any chip to edit**
- FR-01-13 Chips animate onto screen with 80ms stagger and spring physics (scale 0→1.0, opacity 0→1)
- FR-01-14 Mandatory fields: category (with fallback selector), primary colour — user cannot save without these
- FR-01-15 Optional fields: brand, secondary colour, material, pattern, season, notes
- FR-01-16 Store both original AI output and user-corrected version for accuracy measurement and model retraining

Save & First-Outfit Prompt

- **FR-01-17 On save: check if new item unlocks any outfit combinations; if yes, show "This item unlocks 3 new outfits" modal**
- FR-01-18 After save: offer "Scan another item" (primary) or "Done" (secondary) — keeps user in scan flow for batch sessions
- FR-01-19 Batch mode: queue up to 10 photos, process asynchronously, show summary review screen listing all queued items
- FR-01-20 If AI detects coordinated set (matching co-ord, suit), prompt: "Save as a set" or "Save as individual items"

1.4 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-01-01	Scan-to-save completes without error	End-to-end test, 100 items	>98% success rate
AC-01-02	AI category prediction correct	Validation set, 200 images (incl. 40 African garments)	>85% accuracy
AC-01-03	Colour detection matches human label	50-image human benchmark	>90% agreement
AC-01-04	Round-trip latency photo → tags shown	Instrumented test on 3G network	p95 < 10 seconds
AC-01-05	Low-confidence prompt triggers	Unit test on confidence values <60%	100% of sub-60% cases
AC-01-06	User can edit any AI-generated tag	Manual UI test	All fields editable
AC-01-07	Batch scan: 10 items, no crash	Stress test	0 crashes
AC-01-08	Background removal: clean result	Visual QA, 30 images (hanger/flat-lay/worn)	>90% clean backgrounds
AC-01-09	Outfit unlock notification fires	Unit test with known unlock scenario	100% of qualifying saves
AC-01-10	Offline queuing works	Airplane mode test during batch scan	Items queued; upload on reconnect

1.5 Edge Cases

Scenario	System Behaviour	User-Facing Message
Poor lighting	Compute brightness; reject below threshold before API call	"This photo is too dark — try near a window."
Blurry image	Laplacian variance check; reject if below threshold	"Photo is blurry — tap to focus and try again."
No garment detected	Vision API returns no clothing label >30% confidence	"We couldn't find a clothing item. Try a plain background."
Multiple items in frame	Detect multiple garment bounding boxes; prompt user to crop or split	"We found more than one item — scan one at a time for best results."
Kente / Ankara print	Vision API may misclassify pattern; show "Pattern" disambiguation chip with African print options	"What type of print is this?" with options incl. African Print
Near-duplicate item	Hash comparison on image content; warn if >85% similar to existing item	"This looks similar to [Item Name] in your wardrobe. Save anyway?"
API timeout	Retry once after 3s; on second failure fall back to manual entry form	"AI tagging timed out — add details manually."
Item on body (worn photo)	Detect face/body in frame; prompt to use flat-lay or hanger photo	"For best results, photograph items on a hanger or flat surface."

1.6 UI Specification

Screen A — Camera Viewfinder

- Background: #0F0F14 (full dark — same energy as splash)
- Viewfinder: full-bleed live camera feed
- Frame overlay: 4 white corner brackets (24px each), defining a 300x360px garment zone
- Lighting indicator: pill at top center — background shifts green/amber/red
- Bottom sheet (fixed, 140px): "Scan Item" CTA (sage green pill, full-width minus 48px margin) + "Choose from library" text link below
- Top-left: back arrow (white, 22px) — exits scan flow
- Haptic: UIImpactFeedbackGenerator.medium on successful capture

Screen B — Processing State

- Captured garment image centered, slightly scaled down (90%), with pulsing sage green ring animation
- Progress label: "Analysing your item..." → "Identifying colours..." → "Almost done..."

- Progress stages update every 2.5s (or real API progress if available)
- Skeleton chips animate in behind progress label — grey placeholder chips hint at what is coming

Screen C — Tag Review

- Item photo: 160x200px rounded card (20px radius), centered top of screen
- Tags section: chips animate in with 80ms stagger, spring physics
- Each chip: Inter 600 13px, #1A1A2E, bg #F5F0E8, border #E5E5E5, radius 99px, padding 8px 16px
- Selected/active chip (being edited): sage green border #7A9E7E, bg #EEF4EE
- "Save to Wardrobe" CTA: full-width, obsidian bg, sage green icon, Inter 700 15px white
- "Scan Another" secondary link below save button

1.7 API Contract

Endpoint	Method	Request Body	Response
POST /api/scan	POST	{ imageBase64: string, userId: string }	{ itemId, tags: { category, colours[], pattern, material, brand, formalityScore, confidence } }
PUT /api/items/:id	PUT	{ category, colours, pattern, material, brand, notes, season[] }	{ item: ClothingItem, outfitsUnlocked: number }
POST /api/items/batch	POST	{ images: base64[], userId: string }	{ batchId, queuedCount, estimatedSeconds }
GET /api/items/batch/:batchId	GET	—	{ status, completed, total, items[] }

1.8 Success Metrics

Metric	Target	Measurement Method
Scan-to-save completion rate	>80%	Event: scan_initiated → item_saved funnel
AI category accuracy	>85%	Rolling validation set, 200 images updated monthly
Average items scanned per onboarding session	>10	Session event aggregate
Time-to-complete first scan	<90 seconds	Instrumented timer from camera open to item saved

Scan abandonment reason distribution	Track all	Funnel drop: poor lighting / no detection / user quit
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2.1 Feature Overview

Field	Detail
Problem Statement	Having a digital wardrobe is worthless without a brain to make sense of it. The outfit engine is WREN's core value delivery — turning a catalogued wardrobe into 5 daily suggestions that feel personal, weather-appropriate, and stylistically coherent.
Solution	A rule-based scoring engine (MVP) that filters eligible items, generates combinations, and scores them against colour harmony, formality match, weather fit, diversity, and style profile alignment. Each suggestion includes a plain-English rationale.
Key Differentiator	Every suggestion is grounded in items the user actually owns. The engine knows laundry status, wear history, and style preferences — making every suggestion feel curated, not generic. The style note teaches fashion intelligence with every recommendation.
V2 Enhancement	ML-enhanced compatibility model (Polyvore dataset approach) + collaborative filtering: "Users with similar style profiles also loved this combination."

2.2 User Stories

Story 1 — Daily Suggestion

As Amara, I want the AI to suggest outfits using items I actually own, filtered by today's weather and my current occasion, so I can make a confident decision in under 30 seconds every morning.

Story 2 — Style Discovery

As Kofi, I want the AI to surface outfit combinations I wouldn't have thought of myself — especially ones that incorporate pieces I haven't worn recently — so my wardrobe feels like it's growing without me buying anything.

2.3 Algorithm Specification

Inputs

Input	Source	Weight in Scoring
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Full wardrobe item metadata	MongoDB — ClothingItem collection	Foundation — all filters applied to this
Current weather	OpenWeatherMap API (lat/lon from device)	High — temperature gate is binary (in/out)
Occasion context	User selection or calendar inference	High — formality filter applied from this
Recent wear history (14 days)	WearEvent collection	Medium — diversity score penalises recent items
Laundry status	ClothingItem.laundryStatus field	Binary filter — dirty/in-wash items excluded entirely
Style profile	User.styleProfile embedded document	Medium — weights all scoring dimensions
Feedback history	WearEvent.rating + outfit dismissals	Medium — implicit preference model updates

Processing — Rule-Based Scoring Engine (MVP)

1. Filter eligible items: exclude dirty/in-wash, apply season filter (tropical: all year), check minimum category availability (must have at least 1 top + 1 bottom OR 1 dress)
2. Generate combinations: top × bottom × optional outerwear layer × shoes × optional accessory
3. Score each combination across 5 dimensions:
 - Colour Harmony Score (0–1.0): complementary = 1.0, analogous = 0.85, neutral-anchor = 0.9, clash = 0.1
 - Formality Match Score (0–1.0): penalise delta >1 on 4-point scale; delta 0 = 1.0, delta 1 = 0.7, delta 2 = 0.3
 - Weather Appropriateness (0–1.0): item fabric weight vs. current temperature; binary waterproofing check if precipitation
 - Diversity Score (0–1.0): penalise items worn in last 7 days (0.5 penalty), last 3 days (0.8 penalty)
 - Style Profile Alignment (0–1.0): weighted average against archetype, colour preference, and formality range from onboarding quiz
4. Composite score = Colour(0.25) + Formality(0.25) + Weather(0.20) + Diversity(0.15) + StyleProfile(0.15)
5. Return top 5 combinations sorted by composite score
6. Generate plain-English style note for #1 result using a template engine: "[Item A] + [Item B] works because [colour rule] — [Item C] adds [dimension benefit]."

2.4 Functional Requirements

- **FR-02-01 Generate 5 outfit suggestions on demand and pre-generate at app open for instant display**
- FR-02-02 Apply all filters (laundry, season, available categories) before scoring
- FR-02-03 Respect colour harmony rules: complementary, analogous, neutral-anchor, monochromatic
- FR-02-04 Respect formality delta rule: casual items not paired with formal; maximum delta of 1
- FR-02-05 Apply weather gate: exclude items inappropriate for current temperature band
- FR-02-06 "Surprise me" mode: apply diversity score multiplier 1.5x to surface unexpected combinations
- FR-02-07 "Swap an item" replaces one slot from next-best candidate list for that category
- FR-02-08 User actions per card: Save / Mark as worn / Dismiss / See alternatives
- FR-02-09 "Why this works" tooltip: expands style note with colour theory explanation and formality rationale
- FR-02-10 Cache pre-generated outfits; invalidate cache when wardrobe or laundry status changes
- FR-02-11 No duplicate outfit in same session
- FR-02-12 If user dismisses 3+ suggestions in a row, trigger "Help us do better" micro-survey (1 question)

2.5 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-02-01	Never suggests dirty/in-wash items	Unit test: mark items dirty; verify absence in suggestions	100%
AC-02-02	Formality delta ≤ 1 for all suggestions	Automated check on 1,000 generated outfits	100%
AC-02-03	Outfit acceptance rate (saved or worn)	Production analytics, 30-day post-launch	>35%
AC-02-04	User-rated outfit quality	In-app rating prompt (weekly sample)	>3.8 / 5.0
AC-02-05	Generation latency (pre-generated)	Time from screen open to cards visible	<800ms

AC-02-06	Generation latency (on-demand)	Time from "Generate" tap to 5 cards shown	<5 seconds
AC-02-07	No duplicate outfit in same session	Unit test: request 10 outfits in sequence	0 duplicates
AC-02-08	"Swap" returns valid replacement	Automated test on 100 swap requests	>95% valid swaps
AC-02-09	Colour harmony score accuracy	Compare algorithm output vs. fashion stylist panel	>80% agreement

2.6 Edge Cases

Scenario	System Behaviour	User-Facing Message
Wardrobe <10 items	Generate partial outfit with empty slot placeholder	"Add [item type] to complete this look →"
All items in laundry	Show "wardrobe at the cleaners" empty state with laundry tracker link	"Everything is in the wash — check your laundry tracker."
3+ consecutive dismissals	Trigger 1-question micro-survey: "What felt off? Style / Colour / Fit / Occasion"	Survey appears as bottom sheet
Wardrobe is all one formality level	Skip formality delta filter; apply other filters normally	No user message — engine adapts silently
Weather API unavailable	Use last known weather; show "Based on recent weather" label on suggestion	"Using recent weather data"
All items worn in last 3 days	Reduce diversity penalty to 0.3 (vs 0.8) to avoid empty results	No user message — engine adapts silently

2.7 Success Metrics

Metric	Target	Measurement
Outfit acceptance rate	>35%	Wear events / suggestions shown
User-rated outfit quality	>3.8 / 5.0	Weekly random sample, in-app rating prompt
Style note comprehension	>70% find it useful	Post-session survey, first 500 users
Swipe-to-dismiss rate	<40%	Dismiss events / suggestions shown
Suggestions featuring unworn items	>60% of suggestions contain ≥1 item not worn in 14 days	WearEvent cross-reference

3.1 Feature Overview

Field	Detail
Problem Statement	Cold-start problem: without user preference data, the outfit engine generates generic suggestions that feel impersonal and drive early churn. The quiz must collect enough signal in <3 minutes to produce meaningful personalisation from the very first outfit suggestion.
Solution	A 6-question style quiz across 5 screens, framed as a premium editorial experience rather than a form. Image-led selections, no dropdowns, "Skip" always visible. Completion reveals a personalised Style DNA profile card — the user's first emotional connection with WREN.
Key Differentiator	The quiz completion screen is a reveal, not a summary. The user sees their Style DNA label, selected preferences as pill chips, and blurred outfit previews that become the carrot for scanning their first items.
Design Reference	Duolingo (gamified progress, one clear action per screen) × Spotify (taste calibration as identity formation)

3.2 Quiz Flow Structure

Screen	Questions	Input Type	Data Collected
Screen 0: Introduction	None — entry moment	"Let's get started" CTA	Opt-in to quiz
Screen 1: Style Identity	Q1: Style archetype	6 cards, multi-select	styleArchetype[]
Screen 1B: Colour Comfort	Q2: Colour palette preference	3 cards, single-select	colorPreference
Screen 2: Lifestyle	Q3: Typical week	4 cards, multi-select	lifestyleType[]
Screen 2B: Climate Zone	Q4: Climate	3 cards, single-select	climateZone
Screen 3: Categories	Q5: Clothing categories worn	Pill tag cloud, multi-select	categoryInclusions[]
Screen 3B: App Goal	Q6: Primary reason for WREN	4 cards, single-select	primaryGoal
Screen 4: Completion	None — Style DNA reveal	"Start Scanning" CTA	Profile creation trigger

3.3 Functional Requirements

- **FR-03-01 Progress indicator: thin sage green line at top of screen, updates per screen**
- FR-03-02 "Skip for now" link: visible on every screen, collects no data for that question, applies neutral defaults
- FR-03-03 Tap-to-select cards with editorial illustrations: scale 1.0→1.03 on selection, sage border appears
- FR-03-04 Multi-select questions: any number of cards selectable; deselect by tapping again
- FR-03-05 Single-select questions: previous selection deselected immediately on new tap
- FR-03-06 Q1 Style Archetypes: Minimalist / Smart Casual / Business / Streetwear / Maximalist / Eclectic
- FR-03-07 Q4 Climate Zone: "Tropical / Hot" card positioned first and visually emphasised for West African users
- FR-03-08 Q5 Category pills: organic wrap layout, pills auto-size to label length; default pre-selected common categories
- FR-03-09 Completion screen: dark background (#1A1A2E), Style DNA label derived from Q1 + Q2 combination, user selections shown as sage pills on a frosted glass card
- FR-03-10 Blurred outfit preview grid visible behind Style DNA card — communicates "this awaits when you scan items"
- FR-03-11 Style Profile editable at any time from Profile screen — re-running quiz invalidates outfit cache
- FR-03-12 Implicit preference model updates from every wear, save, and dismiss event post-onboarding

3.4 Style DNA Label Matrix

Archetype(s) + Colour Preference	Style DNA Label
Minimalist + Neutral	The Quiet Dresser
Minimalist + Balanced	The Considered Minimalist
Smart Casual + Balanced	The Smart Everyday
Business + Neutral	The Polished Professional

Streetwear + Bold	The Bold Urban
Streetwear + Balanced	The Relaxed Creative
Maximalist + Bold	The Expressive Dresser
Eclectic + any	The Style Explorer
Default (skip)	The Open Book

3.5 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-03-01	Quiz completion rate	Production analytics	>75%
AC-03-02	Skip-all completion rate (skips every question and reaches scan)	User test	100% — app fully functional after skip
AC-03-03	Style DNA label assigned for every combination	Unit test on matrix	100% coverage
AC-03-04	Profile editable post-onboarding	Manual UI test	All 6 answers editable
AC-03-05	Outfit cache invalidated after profile edit	Integration test	100%
AC-03-06	Style profile accuracy self-rating	Post-onboarding in-app survey (Day 3)	>4.0 / 5.0
AC-03-07	Screen-by-screen completion funnel	Analytics per screen	<10% drop per screen

4.1 Feature Overview

Field	Detail
Purpose	The daily re-engagement anchor. WREN must give users a reason to open the app every morning. The Morning Brief delivers one confident outfit recommendation with contextual intelligence — weather, occasion, wear history — in a single scroll.
Design Philosophy	One recommendation, not a list. The hero card must feel like a decision, not a starting point for deliberation. Amara needs to get dressed in 30 seconds; the Home screen is engineered around that constraint.
Design Reference	Calm app (contextual daily moment) × Spotify (confident hero recommendation) × Apple Fitness (stat cards with narrative)
Returning Value	Daily. The weather strip, behavioral nudge, and stat cards all change with context — the screen is never identical two days in a row.

4.2 Screen Sections (top to bottom)

Section	Content	Behaviour
Top Bar (fixed)	Contextual greeting + user name + notification bell + avatar	Greeting changes: morning / afternoon / evening
Weather Strip	Temperature + city + brief condition + fabric tip	OpenWeatherMap API; updates on screen open; tap opens full weather detail
Occasion Filter	"Dressing for:" label + 5 horizontal chips: Work / Casual / Evening / Date / Travel	Active chip re-generates hero outfit for that occasion; persists until changed
Hero Outfit Card	Full-bleed flat-lay + outfit name + style note + DNA signals + item thumbnails + Wear CTA	Primary action; see FS-04 UI spec below for full detail
Quick Stats Row	3 cards: Items total / Outfits available today / Unworn count	Tapping Items → Wardrobe Gallery; Outfits → Recommendations; Unworn → Gallery filtered to unworn
Behavioral Nudge Card	Contextual insight from wear data	Rotates from insight pool daily; tap → relevant screen
More Outfits Preview	"More for today" + horizontal scroll of 3 mini outfit cards	Partial 4th card visible; tap any → Recommendations screen

4.3 Hero Outfit Card — Full Specification

- Dimensions: full-width minus 48px horizontal margin; height 380px; radius 28px
- Background: #1A1A2E (Obsidian) — dark card against warm ivory screen creates maximum contrast
- Top 260px: full-bleed editorial flat-lay photograph on warm off-white surface, object-fit: cover
- "Today's Pick" badge: top-left, sage green bg, Inter 700 10px white, radius 99px
- Confidence badges: top-right, frosted glass pills: "Colour Harmony ✓" + "Formality ✓" + "Weather ✓"
- Outfit name: Cormorant Garamond 26px weight 600 #F5F0E8, margin-top 14px from image
- Style note: 3-line plain-English explanation of WHY outfit works — Cormorant Garamond Italic 15px rgba(245,240,232,0.75)
- Item thumbnails: 3–4 circles (36px each, -8px overlap) at card bottom-left
- "Wear This" CTA: sage green pill button, right-aligned, Inter 700 13px white
- Tap card body: expands to full Outfit Detail sheet

4.4 Behavioral Nudge Card Pool

Trigger Condition	Nudge Text	Tap Destination
Item worn 3+ times in last 7 days	"You've worn your [Item] 3 days in a row — here's a fresh alternative →"	Outfit Recommendations, item swap highlighted
Item not worn in 21+ days	"Your [Item] hasn't been out in 3 weeks — 4 new ways to wear it →"	Item detail in Wardrobe Gallery
Low wardrobe utilization	"You've only worn [X]% of your wardrobe this month — explore what you're missing →"	Style Feed, Insights tab
Item recently scanned	"Your new [Item] pairs with [X] pieces already in your wardrobe →"	Outfit Recommendations, new item featured
Weather change	"It's [X]°C today — here are your warmest outfit options →"	Recommendations filtered by weather

4.5 Functional Requirements

- **FR-04-01 Greeting uses device time:** Good morning (<12pm), Good afternoon (12–5pm), Good evening (>5pm)

- FR-04-02 Weather strip fetches on screen open; shows cached data with "Updated Xm ago" label if fetch fails
- FR-04-03 Occasion filter persists across app sessions until user changes it
- FR-04-04 Hero card is #1 result from outfit generation engine for selected occasion + current weather
- FR-04-05 "Wear This" on hero card logs wear event immediately and shows post-wear rating bottom sheet after 3 hours (notification or next app open)
- FR-04-06 Quick Stats calculate in real time from wardrobe data; "Unworn" = items with 0 wear events
- FR-04-07 Behavioral nudge rotates daily from pool of applicable nudges; same nudge not shown twice in 7 days
- FR-04-08 Empty state (0 items scanned): ghosted clothing silhouettes in dashed frame, "Scan your first item to unlock your Morning Brief", progress indicator "0 of 10 items to first outfit suggestion"

4.6 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-04-01	Hero card displays in <800ms on screen open	Performance test (pre-generated cache)	<800ms p95
AC-04-02	Occasion filter change re-generates hero within 3s	Timed user test	<3s p95
AC-04-03	Weather strip reflects current conditions	Compare app vs. device weather widget	>90% accuracy
AC-04-04	Wear CTA logs correct outfit to WearEvent	Integration test	100% correct item IDs logged
AC-04-05	Empty state renders when wardrobe is empty	State test	100%
AC-04-06	Behavioral nudge not repeated within 7 days	Unit test on nudge pool logic	100%
AC-04-07	Home screen DAU open rate	Production analytics	Target: >50% of registered users on any given day by Month 2

5.1 Feature Overview

Field	Detail
Purpose	Wear logging is the single most valuable data action in WREN. Every logged wear event feeds the outfit engine, populates analytics, updates outfit potential scores, and triggers the post-wear rating flow that provides the richest feedback signal available.
Design Philosophy	One tap. No form. The cost of logging a wear must be zero friction. The "Worn Today" button is available from three entry points: hero card, outfit detail, and item detail. Tapping it is all that is required.
Post-Wear Rating	A 3-outcome rating system replaces star ratings. Three cards — Got compliments / Felt great / Wear again — provide richer algorithm signal than any 1–5 scale while taking no more cognitive effort.

5.2 Functional Requirements

Wear Logging

- **FR-05-01 "Worn Today" / "Log a Wear" available from: Home hero card, Outfit detail, Item detail, Weekly planner**
- FR-05-02 Tap logs WearEvent with: outfitId, item IDs, timestamp, current weather (from cache), current occasion
- FR-05-03 Wear count on each item increments immediately and is visible on item card
- FR-05-04 Last worn date updates on item record
- FR-05-05 Item laundry status changes to "Worn / Needs wash" after logging (if user has auto-laundry-tracking enabled)
- FR-05-06 Wear confirmation: brief sage green checkmark animation (300ms) on the button tapped — feels satisfying

Post-Wear Rating

- **FR-05-07 Post-wear rating bottom sheet triggers 3 hours after a wear event is logged (silent notification or next app open)**
- FR-05-08 Sheet presents 3 reaction cards in equal-width row: "Got compliments 🗨️" / "Felt great ✓" / "Wear again 🔄"
- FR-05-09 Multiple selections permitted: user may tap more than one card

- FR-05-10 Selected card: sage green border + tint bg + scale 1.03
- FR-05-11 "Skip" text link below cards — always visible, no guilt
- FR-05-12 Each selection stores a RatingEvent: { wearEventId, outcome: "compliments"|"feltGreat"|"wearAgain", timestamp }
- FR-05-13 Rating outcomes feed directly into outfit engine preference model with weights defined in FS-02

5.3 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-05-01	Wear log completes in one tap from all three entry points	Manual UI test	100%
AC-05-02	WearEvent record created with correct item IDs	Integration test	100%
AC-05-03	Wear count increments on item card immediately	UI state test	100%
AC-05-04	Post-wear sheet triggers within 5 minutes of 3-hour window	Timing test	>95%
AC-05-05	Rating completion rate	Production analytics	Target >50% of wear events rated
AC-05-06	Multi-selection works correctly (2+ outcomes)	Unit test	100%
AC-05-07	Wear logging works offline (queued)	Airplane mode test	Events queued; upload on reconnect

6.1 Feature Overview

Field	Detail
Purpose	WREN's primary navigation system. A deliberate departure from the iOS/Android default tab bar: a constrained-width floating pill that communicates navigation is a tool you pick up, not a bar anchored to the screen. Used in every session, dozens of times.
Architecture	Two distinct layers: (1) The Pill — a 300px wide obsidian pill housing 4 destination tabs with a pill-within-pill active state. (2) The Camera FAB — a sage green squircle button floating 10px above the pill, the only accent-colored element in the navigator.
Design Reference	Linear (precision tool feel) × Craft app (pill-within-pill active state) × Spotify (dark pill on light background)

6.2 Pill Tab Bar Specification

Property	Value	Rationale
Width	300px	Constrained width creates floating object quality vs. full-width bar
Height	64px	Comfortable tap target with visual weight
Border Radius	9999px (full pill)	Defining visual signature
Background	#1A1A2E (Obsidian)	Maximum contrast against warm ivory screen
Bottom Margin	32px from screen edge	Floats above content; does not anchor to bottom edge
Shadow	0 8px 32px rgba(26,26,46,0.20) + 0 2px 8px rgba(26,26,46,0.12)	Two-layer shadow for depth and definition
Border	1px solid rgba(255,255,255,0.06)	Subtle definition; adds premium quality at close range

6.3 Four Destination Tabs

Tab	Icon	Active Label	Inactive Icon Opacity
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Home	Phosphor: house-simple	"Home"	35%
Feed	Phosphor: squares-four	"Feed"	35%
Closet	Phosphor: hanger	"Closet"	35%
Profile	Phosphor: user-circle	"Profile"	35%

Active Tab State — Inner Pill

- Container: height 40px, width dynamic (icon+label+28px padding), bg #F5F0E8, radius 99px, centered vertically in pill
- Icon: 18px, color #1A1A2E
- Label: Inter 600 12px #1A1A2E, 6px left of icon
- Width expansion animation: icon-only → icon+label, duration 300ms, spring easing
- Icon color transitions before pill expands: inactive rgba → #1A1A2E in 100ms, then pill expands in 200ms

6.4 Camera FAB Specification

Property	Value	Rationale
Shape	64x64px, border-radius 20px (squircle)	Distinct from both full circle FAB and square buttons — unique signature
Background	#7A9E7E (Sage Green)	Only accent color in navigator — unmissable entry point
Icon	Phosphor camera, 26px, white, regular weight	Outline style is more elegant than filled at this scale
Shadow	0 8px 24px rgba(122,158,126,0.45) + 0 2px 8px rgba(122,158,126,0.25)	Green-tinted glow — button feels like a light source
Position	Centered horizontally, 10px above pill top edge	Bridges content and navigation worlds
Idle Animation	Scale 1.0→1.03→1.0, 2.5s loop, ease-in-out	Breathing pulse — subconscious invitation to tap
Pressed State	Scale 0.92, bg #5A7E5E, shadow compressed	Physical press feel

6.5 Functional Requirements

- **FR-06-01 Pill floats 32px above bottom screen edge; does not extend to screen edges**
- FR-06-02 Camera FAB floats 10px above pill top edge; renders above pill in z-index
- FR-06-03 Active tab inner pill expands from icon-only to icon+label on selection with spring animation
- FR-06-04 Inactive tab icons at 35% opacity (rgba(245,240,232,0.35))
- FR-06-05 Notification badge: 8px dusty rose circle, top-right of icon, 1.5px obsidian border
- FR-06-06 Navigator hides on scroll down, returns on scroll up or pause
- FR-06-07 Navigator always visible on Home screen and when scrolled to top of any screen
- FR-06-08 Camera FAB tap: button expands to 200px pill showing "× Scan Item [camera icon]" before launching scan flow
- FR-06-09 Camera and pill animate as single unit during scroll hide/show

6.6 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-06-01	Navigator floats at correct position on all device sizes	Visual test: iPhone SE, 14, 14 Pro Max	100%
AC-06-02	Active tab inner pill expands at 60fps	Reanimated performance profiler	60fps on iPhone 12+; 30fps minimum on lower-end devices
AC-06-03	Camera FAB tap opens scan flow	Manual test	100%
AC-06-04	Navigator hides on scroll down	Scroll event test	100%
AC-06-05	Notification badge renders correctly at all counts	Visual test: 0, 1, 9, 10, 99	100%
AC-06-06	Haptic feedback fires on tab switch	Device physical test	100%

7.1 Feature Overview

Field	Detail
Purpose	Daily discovery and fashion education. A vertically scrolling masonry grid that blends four content types into a feed that is personalised to the user's wardrobe data, making it as engaging as social media but without the social comparison of Instagram.
Unique Position	Unlike Pinterest (external content) or Instagram (other people), the Style Feed is about the user's own wardrobe. "Your Combinations" cards are generated from items the user owns. Intelligence cards are derived from their own wear data. Every card is personally relevant.
Design Reference	Pinterest (masonry layout rhythm) × Spotify Discover Weekly (personal intelligence) × Kinfolk magazine (editorial quality)

7.2 Four Content Types

Type	Frequency	Width	Purpose	Refresh Rate
Your Combination	40%	Single column (174px)	AI outfit from user's own items — discovery mode	Daily (new combos each session)
Style Intelligence	20%	Full width (breakout)	Personalised insight from wear data — dark obsidian card	Weekly (new insight each week)
Fashion Principle	20%	Single or full width	Educational with connection to owned items	Evergreen (cycles through library)
Monthly Wrapped	1/month	Full width (pinned)	Shareable achievement card	Monthly

7.3 Card Specifications

Card Type 1 — Your Combination

- Single column width (174px), height 220–280px (variable based on content)
- Top image (70%): editorial flat-lay of user's items on warm off-white surface
- "Your Style" badge: top-left, sage bg, Inter 700 9px white

- Bottom panel (30%): outfit name (Cormorant Garamond 16px #1A1A2E) + style note (Inter 400 11px #6B6B6B)
- Item DNA dots: 3 colored circles (8px each) showing outfit's color palette
- Bookmark icon: top-right, Phosphor bookmark, 16px; filled state sage green + bg tint

Card Type 2 — Style Intelligence

- Full width, 160px height, #1A1A2E background
- 4 variations: Power Piece / Color Story / Wear Gap / Utilization (each with unique icon + headline)
- 64x64px icon block (left) + headline (Cormorant Garamond 20px #F5F0E8) + body (Inter 400 12px rgba(245,240,232,0.60))
- "See breakdown →" CTA: Inter 500 12px #7A9E7E, bottom-right

Card Type 3 — Fashion Principle

- Single or full width, 300–340px height; educational content always connected back to user's owned items
- Top: illustrated graphic (flat design, 2–3 colors, 180px height)
- Bottom: principle name (Cormorant Garamond 22px #1A1A2E) + description (Inter 400 13px #6B6B6B)
- Connection CTA: "Try this with your [specific owned items] →" — links to outfit using those items

7.4 Functional Requirements

- **FR-07-01 Masonry 2-column grid, 10px gap, variable card heights**
- FR-07-02 Filter chips at top: For You (default) / My Combinations / Style Tips / Insights
- FR-07-03 Filter selection shuffles feed with 200ms fade transition
- FR-07-04 Cards fade-in on scroll entry: opacity 0→1, translateY 16px→0, 300ms ease-out
- FR-07-05 Full-width cards have 0.5x parallax scroll effect
- FR-07-06 Save card to named collection: bottom sheet with collection selector
- FR-07-07 Tapping Your Combination card → opens full outfit detail sheet with Wear/Save CTAs
- FR-07-08 Tapping Intelligence card → opens expanded analytics panel for that insight
- FR-07-09 Fashion Principle CTA taps → opens outfit pre-built with those specific items

- FR-07-10 Feed refreshes: daily (new combinations), weekly (new intelligence card), evergreen (principles cycle)
- FR-07-11 Empty state for <5 items: locked blurred card previews, progress bar "3/5 items to unlock your feed", Fashion Principle cards available immediately (they require no wardrobe data)

7.5 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-07-01	Masonry grid renders correctly on iPhone SE, 14, and Pro Max	Visual test on 3 sizes	100%
AC-07-02	Feed session length	Analytics	Target >3 minutes average
AC-07-03	Card save rate	Save events / feed cards shown	Target >15%
AC-07-04	Intelligence cards use real user wardrobe data	Integration test with known wardrobe	100% data accuracy
AC-07-05	Filter selection transitions at 60fps	Performance profiler	60fps p95
AC-07-06	Empty state renders for <5 items	State test	100%

8.1 Feature Overview

Field	Detail
Purpose	A creative workspace for exploring, discovering, and understanding the wardrobe. Three view modes serve different mental models: Items (browsing individual pieces), Outfits (browsing combinations), Collections (browsing curated groups). The daily return hook is the Spotlight Card.
Key Innovation	Variable-height grid in Items View. Items have different card heights based on category (Dresses: 260px, Tops: 200px, Shoes: 180px, Accessories: 160px). This creates an editorial magazine rhythm vs. the rigid uniform grid every other wardrobe app uses.
Design Reference	Apple Photos (familiar organisation) × VSCO (editorial grid quality) × Notion (flexible view modes)

8.2 Three View Modes

Mode	Grid Layout	Key Feature	Who Uses It
Items View (default)	Variable-height 2-column	Outfit potential badge + wear heat bar on every card	All users — primary wardrobe browsing
Outfits View	Uniform 2-column, 280px height	Outfit DNA pills: formality, colour harmony, weather	Fashion-forward users thinking in outfits
Collections View	2-column card grid with 2x2 collage thumbnails	User-named groups with item count + last updated	Power users, high engagement

8.3 Item Card Anatomy

- Image area (70% of card height): clothing photo on warm off-white background, object-fit contain
- Outfit DNA dots: 3 colored circles (8px) at image bottom showing item's detected colour palette
- Outfit Potential badge: top-right, dark (#1A1A2E) bg, sage text "8 outfits", Inter 700 10px
- Wear Heat badge: top-left, flame icon only (Phosphor flame, 12px), shown only if >5 wear events

- Bottom info strip: item name (Inter 600 13px #1A1A2E) + category dot + brand (Inter 400 11px #6B6B6B)
- Wear Heat Bar: 3px bar at very bottom of card; sage green (>10 wears), dusty rose (4–9), barely visible (1–3), empty with dusty tint (0 — silent alert without the word "unworn")

8.4 Daily Spotlight Card

A full-width dark (#1A1A2E) card at the top of the Items View grid. The AI surfaces one item to rediscover each day. Structure: 64x64px item thumbnail (left) + "Today's Spotlight" overline badge (sage) + item name + context line "Not worn in 18 days · 4 new outfit ideas →". Tap → item detail sheet with pre-built outfit suggestions featuring that item. This single card answers "why would I open the gallery today?" — the most important retention question for any secondary screen.

8.5 Functional Requirements

- **FR-08-01 View mode switcher: 3 tabs (Items / Outfits / Collections) below gallery header**
- FR-08-02 Items View: variable-height grid, heights determined by item category
- FR-08-03 Spotlight Card renders at grid top, daily refresh, AI-selected based on recency + outfit potential
- FR-08-04 Category filter chips horizontal scroll: All + each category with count
- FR-08-05 Smart sort options: Most Versatile / Needs Attention / Recently Added / Seasonal Relevance
- FR-08-06 Search: natural language ("blue casual shirt", "formal shoes") returns tagged results
- FR-08-07 Long-press item card: quick action sheet (Mark Worn / Edit Tags / Add to Collection / Delete)
- FR-08-08 Item detail: bottom sheet, 120x150px item image + outfit potential + wear intelligence + 8-week bar chart + "Works in these outfits" horizontal scroll
- FR-08-09 Outfits View: "Never tried" badge (dusty rose tint) on combinations not yet worn
- FR-08-10 Collections View: "Create New" card with dashed border + plus icon
- FR-08-11 Wear Heat Bar values calculated relative to user's most-worn item (not absolute scale)

8.6 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-08-01	Variable-height grid renders correctly on all device widths	Visual test: SE, 14, Pro Max	100%
AC-08-02	Spotlight Card changes daily	Date-change test	100%
AC-08-03	Outfit potential score is accurate	Cross-reference with outfit engine results for 5 test items	>95% accuracy
AC-08-04	Search returns relevant results	User acceptance test: 10 natural language queries	>80% relevant first result
AC-08-05	Wear heat bar reflects correct wear count	Integration test with known wear events	100%
AC-08-06	Item detail sheet loads in <500ms	Performance test	p95 <500ms
AC-08-07	Gallery return rate (daily opens)	Production analytics	Target >40% of daily active users

9.1 Feature Overview

Field	Detail
Purpose	Dedicated outfit discovery screen serving two modes simultaneously: Decisive Mode (30-second morning decision) via the hero card, and Exploratory Mode (Sunday planning, creative browsing) via the weekly planner and occasion sections.
Core Principle	Confidence is a feature. The hero card is not "option 1 of 5" — it is THE recommendation. The Tweak System below the hero card turns passive consumption into creative participation without undermining the confidence of the primary suggestion.
Design Reference	Tinder (confident single card) × Spotify (curated sections with clear purpose) × Apple Weather (decisive number, detail below)

9.2 Screen Sections

Section	Purpose	Mode Served
Sticky Header	Screen title + filter icon + weather chip	Both
Weekly Planner Strip	7-day horizontal, outfit thumbnails or empty slots	Exploratory — Sunday ritual hook
Hero "Wear This Today" Card	Single dominant recommendation, 460px height, dark bg	Decisive — primary morning use case
Outfit DNA Confidence Row	3 mini pills: Colour ✓ / Formality ✓ / Weather ✓	Both — validates recommendation
Style Note	3-line plain-English WHY explanation	Both — teaches fashion intelligence
"Tweak This Outfit" System	3 swap rows with alternative thumbnails from wardrobe	Exploratory — creative participation
Outfit Memory Banner	Repetition nudge from wear history	Both — behavioral intelligence
Occasion Browsing Rows	"For Work This Week" + "For the Weekend" horizontal scrolls	Exploratory — contextual discovery

9.3 Hero Card Specification

- Height: 460px; background: #1A1A2E; radius: 28px; shadow: 0 16px 64px rgba(0,0,0,0.20)
- Top 280px: full-bleed flat-lay photograph with gradient overlay (transparent → rgba(26,26,46,0.60))
- "Wear This Today" badge: top-left, sage bg, "★ Wear This Today" Inter 700 10px white
- Outfit DNA confidence pills: top-right, 3 frosted glass pills, each with tick mark
- Bottom 180px: outfit name (Cormorant Garamond 26px #F5F0E8) + occasion pill + 3-line style note
- Item thumbnails: 4 circles (40px, -6px overlap) bottom-left; tapping opens that item detail
- Action buttons: Share (outline) + Save (outline) + "Wear Today" (sage green filled, flex-2)

9.4 Tweak System Specification

- Three swap rows below hero card, 12px gap between rows
- Each row: current item thumbnail (52x52px, sage border) + "Swap [category] →" label + 3 alternative thumbnails (44x44px)
- Tapping alternative: selected item gets sage border, hero card image updates to show new combination
- Row 3 includes "+ Add layer" button (dashed sage circle, 44px) for adding outerwear
- Alternatives sourced from wardrobe items in same category, next-best scoring candidates

9.5 Post-Wear Rating — Bottom Sheet

Triggered 3 hours after "Wear Today" tap, or on next app open if app was closed. Three equal-width reaction cards (height 80px each):

- "Got compliments 🗨️" — social performance signal → algorithm prioritises for social contexts
- "Felt great all day ✓" — comfort/confidence → algorithm prioritises similar fabric/formality combos
- "Wear again 🔄" — repeat intent → algorithm surfaces combination more frequently

9.6 Functional Requirements

- **FR-09-01 Hero card displays #1 outfit engine result for current occasion and weather**
- FR-09-02 Weekly planner strip: tap empty day slot → opens outfit picker for that day; tap filled slot → opens that outfit
- FR-09-03 Tweak system updates hero card composition when user swaps an item
- FR-09-04 Outfit Memory Banner: shows if a near-identical combination was worn within 14 days; suggests an alternative
- FR-09-05 Occasion browsing rows: "For Work" shows formality 3–4 outfits; "For the Weekend" shows formality 1–2
- FR-09-06 Browse cards show "Worn last [day]" memory badge if outfit worn within 14 days
- FR-09-07 Swipe right on hero card: marks as worn; swipe left: dismisses and shows next recommendation

9.7 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-09-01	Hero card shows correct top-ranked outfit	Integration test with known wardrobe	100%
AC-09-02	Tweak swap updates hero image	UI state test	100%
AC-09-03	Weekly planner persists assignments	State persistence test	100%
AC-09-04	Outfit Memory Banner triggers correctly	Unit test with known wear history	100% of qualifying cases
AC-09-05	Swipe gestures work correctly	Gesture test on 3 device sizes	100%

10.1 Feature Overview

Field	Detail
Purpose	The feature users screenshot and share. An AI-composed editorial flat-lay of any outfit from the user's wardrobe — items arranged on a warm off-white surface as if shot by a fashion photographer. Every result is shareable branded content.
Growth Engine	Each exported flat-lay carries the WREN watermark into social media, group chats, and WhatsApp. Users sharing their Wren flat-lay is free marketing at exactly the right demographic. The share rate is a direct measure of product quality.
Design Reference	Acne Studios lookbook × Kinfolk editorial flat-lay photography × Apple product photography lighting principles

10.2 Flat-Lay Studio Flow

7. Entry: user taps "Try On" from outfit card, or "Studio" CTA in Recommendations screen
8. Mode Selection: Mode Selection screen — two cards: "Flat-Lay Studio" (recommended, dark) and "Try It On" (on-body, light)
9. Canvas: 380px canvas on warm off-white background with 4 item cards (PNG, no background) floating in editorial arrangement
10. Controls: canvas toolbar (arrange / swap / remove / AI enhance) as floating pill at canvas bottom
11. Item selector: horizontal chip strip below canvas; "Add item +" chip for adding more pieces
12. Generate: "Generate Flat-Lay" CTA — obsidian, full-width, shimmer animation indicating AI readiness
13. Result: dark background reveal screen with generated flat-lay + Outfit DNA panel + Save/Wear/Share actions
14. Export: 1080x1080px Instagram-square export with outfit name, item names, WREN watermark

10.3 Canvas Specification

- Canvas surface: #FAF7F2 (warm off-white), subtle linen texture overlay at 3% opacity
- Item cards: PNG with no background (background removal applied at scan time), soft drop shadow
- Default editorial arrangement: shirt top-left, trousers diagonal center, shoes bottom-right, accessory top-right
- Item overlap at natural connection points: shirt over trouser waist — creates outfit silhouette
- Selected item state: 2px sage green selection ring, 4px offset, corner resize handles
- Rotation pre-applied per category: shirt -3°, trousers 2°, shoes 4°, watch -5° — adds organic editorial feel
- Canvas toolbar: floating obsidian pill, backdrop-blur 12px, 4 tool icons at 32x32px hit targets

10.4 Result Screen

- Background: #0F0F14 (same dark as splash — this is a reveal moment)
- Generated flat-lay: 420px tall, radius 28px, shadow 0 16px 64px rgba(0,0,0,0.40)
- WREN watermark: bottom-right of image, Cormorant Garamond Light 11px rgba(26,26,46,0.20) — present but unobtrusive
- Outfit DNA panel: below image, 3 dimension bars (Colour Harmony / Formality / Weather) + style note in Cormorant Garamond Italic
- Action buttons: "Save to Collection" (outline) + "Wear Today" (sage green, flex-2)
- "Share this look" text CTA: opens export card at 1080x1080px

10.5 Shareable Export Card Specification

- Dimensions: 1080x1080px (Instagram square format)
- Background: #FAF7F2 (warm off-white — same as canvas surface)
- Top 60%: flat-lay composition, centered, taking dominant space
- Bottom 40%: outfit name (Cormorant Garamond 28px #1A1A2E) + item names row (Inter 300 13px #6B6B6B) + WREN bird mark + wordmark
- WREN branding: bird mark (30px stroke) + "WREN" wordmark Cormorant Garamond Light 14px, letter-spacing 6px
- Result looks like a fashion brand's Instagram post — users want to share it without prompting

10.6 Functional Requirements

- **FR-10-01 Background removal must have been applied to all item images at scan time for canvas PNG display**
- FR-10-02 "AI Enhance" canvas tool calls server-side composition algorithm to optimise item positions editorially
- FR-10-03 "Swap item" opens wardrobe category picker filtered to same category
- FR-10-04 Generate button triggers server-side flat-lay composition via Sharp.js pipeline
- FR-10-05 Result image stored in user's try-on history for 30 days
- FR-10-06 Export card renders at 1080x1080px; share sheet opens on tap
- FR-10-07 Processing fallback: if generation takes >10s, show progress messaging with wardrobe insight

10.7 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-10-01	Flat-lay generation succeeds for 4-item outfit	Integration test with 10 different outfit combinations	>95% success rate
AC-10-02	Generation latency	Timed test (server-side composition)	p95 <8 seconds
AC-10-03	Export card renders at 1080x1080px	Image dimension verification	100%
AC-10-04	Share rate from try-on screen	Production analytics	Target >20% of try-on sessions
AC-10-05	WREN watermark present on all exports	Automated image inspection	100%
AC-10-06	Background removal clean (no background artifacts)	Visual QA on 30 item images	>90% clean

11.1 Feature Overview

Field	Detail
Purpose	The Sunday ritual retention hook. Users who plan Monday's outfit on Sunday have a measurably higher probability of opening the app on Monday morning. This weekly planning behavior creates a commitment loop that is the strongest predictor of long-term retention.
Design Reference	Fantastical (clean calendar UX where content is the interface) × Google Calendar (drag-and-drop simplicity)
Behavioral Mechanic	A user who has planned 3 days' outfits on Sunday has created 3 future reasons to open the app. Each planned day is a micro-commitment. The planner is not a calendar feature — it is a retention engine.

11.2 Functional Requirements

- **FR-11-01 7-day horizontal strip (Mon–Sun): each day shows date number + day abbreviation + 36px circle**
- FR-11-02 Empty day circles: dashed border, plus icon; tap → opens outfit picker for that day's weather and calendar events
- FR-11-03 Assigned day circles: outfit thumbnail fills circle, sage green ring border
- FR-11-04 TODAY: dark pill background, sage day label, ring indicator — visually distinct from other days
- FR-11-05 Drag-and-drop: drag outfit card from browsing section onto a day circle in strip
- FR-11-06 Device calendar integration (read-only): event names shown below day as context chips
- FR-11-07 Repeat detection: warn if same outfit planned within 14-day window for tagged social events
- FR-11-08 Lock outfit: lock icon prevents AI from suggesting changes to locked days
- FR-11-09 Planned outfits auto-log wear event when date passes, or prompt "Did you wear this?" on the morning
- FR-11-10 Sunday evening push notification: "Plan your week in 5 minutes →" — deep links to planner

11.3 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-11-01	Outfit assignment persists across sessions	State persistence test	100%
AC-11-02	Repeat outfit warning triggers correctly	Unit test: same outfit 3 days apart, 13 days apart	100% within 14-day window
AC-11-03	Calendar integration read-only	Permission audit	No calendar write events
AC-11-04	Lock prevents AI suggestion override	Integration test	100%
AC-11-05	Sunday reminder delivered	Push notification delivery test	>95% delivery rate
AC-11-06	Weekly plan creation rate	Production analytics	Target >30% of active users complete a weekly plan by Month 2

12.1 Feature Overview

User-named saved outfit groups. Collections turn the wardrobe gallery from a flat list into a personally organised creative system — "Work Week", "Going Out", "Travel Capsule". Inspired by Notion's approach to user-defined organisation: flexible, user-controlled, powerful without complexity.

12.2 Functional Requirements

- **FR-12-01 Create collection: tap "+ Create New" card in Collections view → name input sheet**
- FR-12-02 Add outfit to collection: from any outfit card (feed, recommendations, gallery) via bookmark → sheet with collection selector
- FR-12-03 Collection card: 2-column grid, 180px height, 2×2 collage of outfit thumbnails, name, "8 outfits · Updated 2d ago"
- FR-12-04 Within collection: outfit grid with horizontal scroll, occasion filter, sort by date/wear frequency
- FR-12-05 Outfit engine filter: option to generate suggestions from a specific collection only ("Suggest from my Work Capsule")
- FR-12-06 Free tier: maximum 2 collections; Pro tier: unlimited
- FR-12-07 Delete collection: swipe-to-delete with confirmation; outfits remain in wardrobe

12.3 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-12-01	Collection creation and naming flow	User acceptance test	100% completion
AC-12-02	Outfit engine correctly filters to collection items	Integration test	100% of suggestions from collection items only
AC-12-03	Free tier collection limit enforced	State test: attempt to create 3rd collection	Paywall shown on 3rd creation attempt

AC-12-04	Collection card thumbnail collage renders correctly	Visual test	100%
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13.1 Feature Overview

Field	Detail
Purpose	A shareable monthly achievement card that celebrates each user's personal style data in a beautiful, story-driven format. The word-of-mouth growth driver for WREN — each shared card is branded WREN content reaching the right demographic organically.
Design Reference	Spotify Wrapped — the defining proof that personal data can be emotionally resonant and socially shareable when told as a story.

13.2 Wrapped Card Content

Metric	Display Format	Example
Items worn	Large number + label	"32 items worn"
Outfits created	Number + label	"19 new combinations"
Wardrobe utilization	Percentage + percentile vs. WREN users	"68% utilization · top 15% of WREN users"
Most-worn item	Item thumbnail + name + wear count	"Your go-to: White Oxford (12 wears)"
Outfit DNA this month	Colour palette swatches derived from worn items	Visual colour story — 3–5 hex swatches
Achievement unlock	Badge if threshold met	"Wardrobe Explorer — 50+ items scanned"

13.3 Functional Requirements

- **FR-13-01** Wrapped generates on the 1st of each month for previous month's data
- **FR-13-02** End-of-month push notification: "Your [Month] Wrapped is ready →"
- **FR-13-03** Card rendered as full-screen animated sequence (5 screens, tap to advance)
- **FR-13-04** Final screen: static shareable image at 1080x1920px (Instagram Story format) with WREN branding
- **FR-13-05** Wrapped card pinned to top of Style Feed for the first week of the month
- **FR-13-06** Share rate tracked as primary success metric

13.4 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-13-01	Wrapped generates correctly for previous month's wear data	Integration test with 3 known wear histories	100% correct metrics
AC-13-02	Share rate	Production analytics	Target >40% of monthly active users share their Wrapped
AC-13-03	Export renders at 1080x1920px	Image dimension verification	100%
AC-13-04	Push notification delivered	Delivery test	>95% delivery rate

14.1 Feature Overview

Field	Detail
Purpose	The showstopper feature. AI compositing of an outfit onto a user-provided photo. This is the feature users tell their friends about: "It literally showed me wearing my own clothes differently." Drives premium tier conversion and social sharing at higher rates than flat-lay.
Approach	Photo-based (not real-time AR). User uploads a photo; AI composites outfit onto it and returns 3 variations (e.g. shirt tucked/untucked/jacket added). More convincing than real-time AR (no motion consistency required) and achieves the core goal: seeing yourself in the outfit.
Privacy	Photos processed server-side via AI compositing API; photos never stored permanently; processed and deleted within 24 hours; privacy notice shown before photo upload. On-device processing explored for V3.

14.2 On-Body Try-On Flow

15. Entry from Mode Selection screen (FS-10 Frame 1): tap "Try It On" card
16. Photo source: "Take a photo now" (camera) or "Choose from library"; guidance overlay shows ideal pose (standing, neutral background)
17. Processing screen: split composition (user photo left, scanning animation right); wardrobe insight displayed during wait
18. Result screen: user photo with outfit composited; variation selector strip at image bottom (3 thumbnails)
19. Comparison toggle: "Try-On" / "Original" — reveals the original photo for direct comparison
20. Outfit DNA panel and action buttons identical to Flat-Lay result (FS-10)

14.3 Functional Requirements

- **FR-14-01 Photo capture with pose guidance overlay: arrow indicating standing position, background tip**
- **FR-14-02** Submit to AI compositing API (e.g. outfit transfer model); receive 3 variation images

- FR-14-03 Variation labels: contextual to outfit (Tucked In / Casual / + Jacket)
- FR-14-04 "Original" toggle side-by-side comparison mode
- FR-14-05 Photo deletion: user photo deleted from server within 24 hours post-processing; deletion confirmed in privacy notice
- FR-14-06 Privacy notice: explicit consent screen before photo upload
- FR-14-07 Processing takes 8–15s; "Did you know?" insight cards cycle during wait
- FR-14-08 Gate behind Pro tier; free users see blurred result with "Upgrade to see yourself in this outfit"

14.4 Acceptance Criteria

ID	Criterion	Test Method	Pass Threshold
AC-14-01	Compositing result visually plausible	User panel evaluation: 20 participants rate 1–5	>3.5 / 5.0 average plausibility
AC-14-02	Processing latency	Timed API test	p95 <20 seconds
AC-14-03	Photo deleted within 24 hours	Server audit log verification	100%
AC-14-04	Privacy consent shown before upload	UI test	100%
AC-14-05	Pro tier gate works correctly	State test	100% free users see upgrade prompt
AC-14-06	Share rate (On-Body vs Flat-Lay)	A/B analytics	On-Body target >30% (vs Flat-Lay >20%)

Non-Functional Requirements

Performance

Requirement	Target	Applies To
App launch to interactive (cold start)	<3 seconds	All screens
Screen navigation transition	<200ms	All tab switches and push transitions
Outfit generation (pre-cached)	<800ms to cards visible	Home, Recommendations
Outfit generation (on-demand)	<5 seconds	Recommendations, refresh
Image load (thumbnails, cached)	<300ms	Gallery, Feed, Home
API response (outfit engine)	<2 seconds p95	Server-side scoring
Scan round-trip (photo → tags)	<10 seconds p95 on 3G	Scan flow
Flat-lay generation	<8 seconds p95	Try-On Studio

Security & Privacy

Requirement	Implementation
User authentication	JWT + refresh tokens stored in expo-secure-store (encrypted keychain)
Data in transit	TLS 1.3 minimum for all API calls
Data at rest	MongoDB Atlas encryption at rest; S3 server-side encryption (AES-256)
Image privacy	Item photos stored in user-scoped S3 prefix; no cross-user image access possible
On-body try-on photos	Processed and permanently deleted within 24 hours; user consent required; no training use
GDPR compliance	Data export (JSON, one-tap); account deletion (all data, permanent, 30-day processing)
Analytics	No PII in analytics events; user ID hashed; opt-out available in Settings

Accessibility

Requirement	Standard	Implementation
Colour contrast	WCAG AA (4.5:1 minimum)	All text/background pairs verified at design token level
Touch target size	Apple HIG (44x44px minimum)	All interactive elements meet minimum
Screen reader support	VoiceOver (iOS) + TalkBack (Android)	accessibilityLabel on all meaningful elements
Dynamic Type	iOS Dynamic Type	Inter and Cormorant respect text size preference
Reduced Motion	iOS reduceMotion preference	All spring animations replaced with opacity-only fades
Keyboard navigation	Full keyboard control	All inputs and CTAs keyboard-accessible

Offline Behaviour

Screen	Offline Behaviour
Home	Show cached outfit from last session; weather shows "cached" label
Wardrobe Gallery	Fully functional from local cache; last-sync timestamp shown
Scan	Photos queued locally; upload resumes when online (background task)
Style Feed	Last 24 hours of feed cached; "You're offline" banner if stale
Outfit Recommendations	Show last pre-generated set; no regeneration until online
Wear Logging	WearEvents queued offline; sync on reconnect

WREN

Wear with intention.

Feature Specification Document v2.0

14 Features · FS-01 through FS-14 · MVP + V1 + V2 Scope

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